



Alberta Public Health Disease Management Guidelines

Legionellosis



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Table of Contents

Case Definition	4
Confirmed Case	4
Probable Case	4
Reporting Requirements	5
Physicians, Health Practitioners and Others	5
Laboratories	5
Alberta Health Services, First Nations and Inuit Health Branch	5
Epidemiology.....	6
Etiology	6
Clinical Presentation.....	6
Diagnosis	6
Treatment.....	6
Reservoir.....	6
Transmission.....	7
Incubation Period	7
Period of Communicability.....	7
Host Susceptibility	7
Incidence.....	8
Public Health Management.....	9
Key Investigation.....	9
Case Management.....	9
Management of Contacts	9
Management of Outbreaks.....	10
Preventive Measures.....	10
Appendix 1: Revision History	11
References.....	12

Case Definition

Confirmed Case

Clinical illness^(A) with laboratory confirmation of infection:

- Isolation of *Legionella* species or detection of the antigen from a lower respiratory tract specimen or a sterile site clinical specimen^(B);

OR

- Detection of *Legionella* nucleic acid in a lower respiratory tract specimen by polymerase chain reaction (PCR);

OR

- Seroconversion or significant (fourfold or greater) rise in *Legionella* sp. IgG antibody titre between acute and convalescent sera;

OR

- Detection of *L. pneumophila* antigen in urine^(C).

Probable Case

Clinical illness^(A) with one of the following:

- Single IgG antibody titre to *Legionella* sp. of > 1:128;

OR

- Epidemiologically linked to a confirmed case.

^(A) Legionellosis comprises two distinct illnesses:

- Legionnaires' disease characterized by myalgia, fever, cough, and
- Pneumonia and Pontiac fever, a milder illness without pneumonia.

^(B) Refer to the [Public Health Laboratory \(ProvLab\) Guide to Services](#) for current specimen collection requirements.

^(C) Cross-reactions have been demonstrated between urinary antigens of several *L. pneumophila* groups. Antigen may be excreted for months after acute infection.

Reporting Requirements

Physicians, Health Practitioners and Others

Physicians, health practitioners and others shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all confirmed and probable cases in the prescribed form by mail, fax or electronic transfer within 48 hours (two business days).

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- Chief Medical Officer of Health (CMOH) (or designate), and
- MOH (or designate) of the zone.

Alberta Health Services, First Nations and Inuit Health Branch

- The MOH (or designate) of the zone where the case currently resides shall forward the initial Notifiable Disease Report (NDR) of all confirmed and probable cases to the CMOH (or designate) within two weeks of notification and the final NDR (amendment) within four weeks of notification.
- For out-of-province and out-of-country reports, the following information should be forwarded to the CMOH (or designate) by phone, fax or electronic transfer within 48 hours (two business days):
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.

Epidemiology

Etiology

Legionellosis is caused by *Legionella* bacteria, which are gram-negative bacilli.⁽¹⁾ There are at least 20 species of *Legionella* that cause disease in humans. The most common type is *L. pneumophila* serogroup 1 and causes the majority of clinical disease.

Clinical Presentation

Legionellosis is an acute bacterial disease with two currently recognized, distinct manifestations: Legionnaires' disease and Pontiac fever.⁽²⁾

Legionnaires' disease is the more severe form of Legionellosis, characterized by pneumonia and progressive respiratory distress that can lead to death. Case-fatality rates are around 15%, even though treatment and diagnostics have been improved.⁽²⁾ Pneumonia caused by *Legionella* cannot be accurately differentiated from pneumonia caused by other organisms.⁽³⁾ Other symptoms associated with the infection include:

- cough,
- fever,
- chills,
- myalgia, and
- gastrointestinal tract, renal and central nervous system manifestations.^(1,4)

Pontiac fever is a milder febrile illness without pneumonia. Deaths due to Pontiac fever have not been reported. Symptoms include abrupt onset of influenza-like illness.⁽¹⁾ Pontiac fever is most often seen as part of a cluster or during outbreaks.^(1,5)

Diagnosis

Laboratory testing for the detection of *Legionella* may include urine antigen testing, serum antibody testing, respiratory culture and other molecular methods (e.g., PCR). The gold standard for diagnosis is respiratory culture, which allows the identification of species and subgroups. Urine antigen testing can also be used; however, it is only highly sensitive and specific for *L. pneumophila* serogroup 1.⁽¹⁾ Refer to [ProvLab Guide to Services](#) for current specimen collection recommendations.⁽¹¹⁾

Treatment

- Consultation with an Infectious Diseases Specialist is recommended.
- Treatment of Legionnaires' disease with appropriate antibiotics is recommended.
- Cases with Pontiac fever generally recover spontaneously in two to five days without treatment.⁽¹⁾

Reservoir

Legionella bacteria are ubiquitous in nature, particularly in aquatic environments.^(1,2) Temperatures of 25–45°C, stagnant water, presence of ciliate protozoans and amoebas, corrosion and biofilms are some of the factors that support growth of *Legionella* bacteria. Scale, which is the build-up of sediment or minerals (calcium or magnesium salts) in or on water distribution systems, household plumbing or water reservoirs also contribute to growth of *Legionella*. Outbreaks and sporadic cases have been linked to:

- hot and cold water systems,
- air conditioning cooling towers and evaporative condensers,

- humidifiers for food display cabinets or home humidifiers,
- whirlpool spas/natural pools/thermal springs,
- respiratory therapy devices,
- decorative fountains/sprinklers,
- garden hoses,
- potable water systems,
- vehicle washes,
- mist producing devices, and
- water-cooled machine tools.^(1,4)

Transmission

Legionella bacteria can be transmitted from water to humans by the inhalation of contaminated aerosols and by aspiration.⁽¹⁾ Person-to-person transmission does not usually occur but may be possible in rare cases.⁽⁶⁾

Incubation Period

The incubation period has a range of two to 14 days and usually averages five to six days.^(4,7,8) Some cases may have an extended incubation period of up to 19 days.

Pontiac fever has a much shorter incubation period with a range of 5–72 hours, but most often 24–48 hours.

Period of Communicability

Unknown; person-to-person transmission is rare.⁽⁶⁾

Host Susceptibility

The chances of developing Legionellosis depends on host factors, production and dissemination of aerosols, concentration of *Legionella* bacteria in the water source, and the virulence of the particular strain of *Legionella*.⁽⁹⁾ Individual risk factors for acquiring disease include:

- older age group (> 50 years),
- male gender,
- heavy smoking or high alcohol intake,
- chronic disease such as diabetes, lung, kidney or liver disease,
- immunocompromising disorders or immunosuppressant medications, and
- heavy smoking.

Risk factors for exposure to *Legionella* include:

- recent travel, with an overnight stay outside of the home, including a stay in a health care facility,
- exposure to hot tubs, and
- exposure to settings where the plumbing has had recent repairs or maintenance work.

Because health care associated infections can occur, Legionnaires' disease should be considered as part of the differential diagnosis in patients who develop pneumonia during or after their hospitalization. Transplant patients, those who are immunocompromised or with certain chronic underlying health conditions are at highest risk of acquiring nosocomial

Legionella infection. Legionellosis is rare in children and if infection does occur, symptoms are usually mild, asymptomatic and may go unrecognized.⁽¹⁾

Incidence

Between 2000 and 2016 there were 106 cases of Legionellosis reported in Alberta.⁽¹⁰⁾ The majority of those cases were acquired locally. In the summer/fall of 2012, there was a reported increase in locally-acquired Legionellosis cases. Intense public health investigations did not reveal a common nosocomial or residential source linking the cases.

Annual case counts can be accessed through the [Interactive Health Data Application](#) (IHDA).

Public Health Management

Key Investigation

- Confirm the diagnosis. Other causes of pneumonia, such as influenza, pneumococcal or streptococcal disease, should be ruled out if possible.
- Assess for any underlying health conditions.
- Determine where the case may have been exposed, taking into consideration the incubation period, reservoir and mode of transmission:
 - Obtain detailed travel history including names of hotels, resorts, etc. during the 14 days prior to symptom onset.
 - History of relevant exposure within the 14 days prior to symptom onset, particularly to air conditioners, humidifiers and aerosolized water (e.g., hot tubs, jacuzzis, fountains, plant misters), grocery stores or cooling stations.
 - Assess if there have been any changes to the main source of water in the primary residence or travel settings, including general construction, plumbing projects, water main breaks, water line work or changes in water temperature during the two to four weeks prior to symptom onset.
 - Conduct an epidemiologic and environmental assessment in consultation with appropriate Environmental Public Health and/or facility-based Infection Prevention and Control personnel to help determine whether environmental sampling is indicated as part of the *Legionella* investigation.
 - If indicated, attempt to collect water samples from potential sources of exposure that may be contaminated; submit to ProvLab for examination (see [ProvLab Guide to Services](#) for current water sample collection requirements).
- Determine if exposure may have occurred in a community health care setting such as a doctor or dentist office/clinic, long-term care facility, assisted/senior living facility, or nursing home.
- Determine if the person was a patient, resident, visitor or worker at the time of exposure in the particular setting.
- If hospitalized, attempt to determine if the exposure was in the community or nosocomial.

Table 1: Classification of Nosocomial Exposure

Classification	Definition
Definite nosocomial exposure	Confirmed case of <i>Legionella</i> who is hospitalized continuously for 14 days before the onset of symptoms
Possible nosocomial exposure	Confirmed case of <i>Legionella</i> who developed symptoms 2–13 days after hospital admission

Management of Cases

- Routine practices are recommended for hospitalized patients.

Management of Contacts

Person-to-person transmission does not usually occur; however, symptomatic and asymptomatic contacts should be investigated if a common source is suspected (e.g., cooling tower).

Management of Outbreaks

- A *Legionella* outbreak is defined as two or more confirmed cases linked by time and space.
- The MOH (or designate) of the zone may declare an outbreak in the manner determined most appropriate, in consultation with Alberta Health.
- For the purposes of public messaging, the term "outbreak" need not be used in order to apply the outbreak policy.
- Outbreak measures include but are not limited to the following.
 - Conduct an epidemiologic and environmental assessment and identify source of exposure.
 - Attempt to collect water samples from potential sources of exposure that may be contaminated and submit to ProvLab for examination (See [ProvLab Guide to Services](#) for current water sample collection requirements).
 - Implement measures to prevent further exposure to identified source of exposure.
 - Review maintenance logs for water systems for sources of exposure.
 - Review and implement decontamination procedures of contaminated water systems.

NOTE: Travel-related *Legionella* outbreaks can sometimes be difficult to identify, especially if cases return home prior to symptom onset.

Preventive Measures

- Proper maintenance of water systems (especially artificial water sources) in which *Legionella* grow is the key to preventing infection with *Legionella*.⁽¹²⁾
- General recommendations for prevention of infection include the following.
 - The elderly and those with immunocompromising conditions should avoid high risk areas such as whirlpool spas.⁽¹³⁾
 - Mist-producing devices such as humidifiers, hot tubs, shower heads and whirlpool bathtubs/Jacuzzi should be cleaned and disinfected as per manufacturer instructions.⁽¹⁴⁾
 - Home water heaters should be kept at a minimum temperature of 60°C; however, in order to prevent scalding, the temperature at the tap should be no higher than 49°C.⁽¹⁴⁾
 - Tap water should not be used for respiratory therapy devices.⁽²⁾ Instead, sterile water should be used in the device tank and for rinsing any type of equipment used for respiratory tract treatment.
 - Keep hot and cold water systems clean and reduce stagnation by flushing unused taps.⁽⁹⁾
 - Cooling towers should be well maintained. Maintenance, testing and disinfection of water-cooling towers is a complex process and should be done by persons with expertise in this area.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	• Updated Template • Diagnosis and Treatment section moved to Epidemiology • Updated web links
	Key Investigation	• Classification of nosocomial exposure put into a table

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